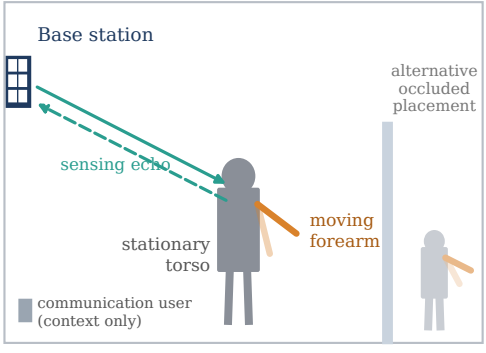
(a) One sensing task



Same room. Same arm motion. Same deadline.

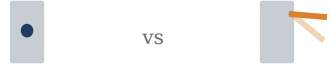
Detect motion onset within 200 ms

3.5 GHz	12 GHz	15 GHz	28 GHz

FR3

(b) Three modeling choices

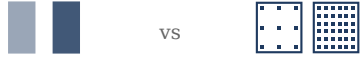
Target



Stationary point: no independent limb motion

Articulated target: motion enters the echo

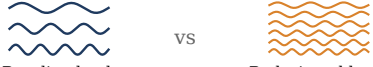
Antenna



Fixed gain (aperture shrinks with f)

Aperture-based gain scaling (idealized comparison)

Detector



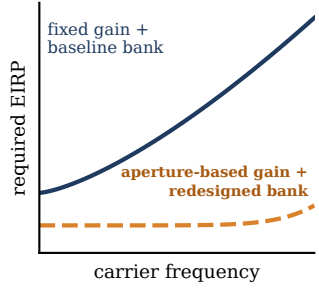
Baseline bank (coarse motion grid)

Redesigned bank (finer motion grid)

(c) The design question

How much sensing EIRP does each band need?

(conceptual illustration, not data)



Interpret a band comparison together with its target, antenna, and detector assumptions.